

Question 1. (7 Points, 2/5 points per part)

A longitudinal study measures a continuous response y_{ij} for subject i at two ordered time points

$$t_j \in \{0, 1\},$$

where $t = 0$ denotes baseline and $t = 1$ denotes follow-up.

Each subject belongs to either Control (0) or Drug (1):

$$\text{drug}_i \in \{0, 1\}.$$

Consider the linear mixed model

$$y_{ij} = \beta_0 + \beta_1 t_j + \beta_2 \text{drug}_i + \beta_3 (t_j \cdot \text{drug}_i) + b_{0i} + \varepsilon_{ij},$$

where

$$b_{0i} \sim N(0, \tau_0^2),$$

and the within-subject residual vector

$$\varepsilon_i = (\varepsilon_{i0}, \varepsilon_{i1})^\top$$

satisfies

$$\text{Var}(\varepsilon_{i0}) = \sigma^2, \quad \text{Var}(\varepsilon_{i1}) = \sigma^2 \delta^2,$$

$$\text{Corr}(\varepsilon_{i0}, \varepsilon_{i1}) = \phi.$$

Assume random effects and residuals are independent.

- (a) Write the Level-1 and Level-2 representations (HLM form).

(b) Let

$$\mathbf{y}_i = \begin{pmatrix} y_{i0} \\ y_{i1} \end{pmatrix}.$$

Find the marginal distribution of $\mathbf{y}_i$.

Your answer must clearly specify:

- the marginal mean vector,
- the marginal covariance matrix,

in terms of $\beta_0, \beta_1, \beta_2, \beta_3, \tau_0^2, \sigma^2, \delta, \phi$.

Question 2. (9 Points ,3/3/3 Points Per Part)

The following model was fit using REML in R:

Linear mixed model fit by REML

Formula: score ~ hours + gender + (1 + hours | school)

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
school	(Intercept)	400	20	
	hours	25	5	0.30
Residual		100	10	

Fixed effects:

	Estimate	Std.Error
(Intercept)	50.0	2.0
hours	3.0	0.5
gender	-4.0	1.2

Additionally, the conditional modes (EBLUPs) for school 5 are:

$$\hat{U}_{0,5} = 10, \quad \hat{U}_{1,5} = -2.$$

- (a) Write the fitted Linear Mixed Model equation, clearly labeling fixed and random parts.

(b) Write the estimated random effects covariance matrix D .

(c) For a student in school 5 with

hours = 8, gender = 1,

write:

- The marginal mean
- The conditional mean

Question 3. (14 Points, 4/1/2/2/5 points per part)

Two coins are tested independently.

- Coin A is tossed $n_A = 4$ times, with $x_A = 4$ heads.
- Coin B is tossed $n_B = 4$ times, with $x_B = 0$ heads.

Assume that X_A, X_B are independent and follow:

$$X_A \sim \text{Binomial}(4, p_A), \quad X_B \sim \text{Binomial}(4, p_B).$$

The joint likelihood is

$$L(p_A, p_B) = \binom{4}{x_A} p_A^{x_A} (1 - p_A)^{4-x_A} \binom{4}{x_B} p_B^{x_B} (1 - p_B)^{4-x_B}.$$

Here $\binom{4}{k} = \frac{4!}{k!(4-k)!}$. We use likelihood ratio test to test

$$H_0 : p_A = p_B \quad \text{vs} \quad H_a : p_A \neq p_B.$$

Answer the following questions.

- (a) Find the Maximum likelihood estimators (MLEs), i.e. the estimated value for p_A, p_B which maximize $L(p_A, p_B)$, without restriction in H_0 .

(b) Find the MLEs under H_0 .

(c) Write the likelihood ratio $\Lambda(x_A, x_B)$ and compute its value for the observed data.

(d) Compute $-2\log \Lambda$. Using the large-sample approximation

$$-2\log \Lambda \sim \chi_1^2,$$

state the decision at $\alpha = 0.05$.

df	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$	$\alpha = 0.001$
1	2.71	3.84	6.63	10.83
2	4.61	5.99	9.21	13.82
3	6.25	7.81	11.34	16.27
4	7.78	9.49	13.28	18.47
5	9.24	11.07	15.09	20.52

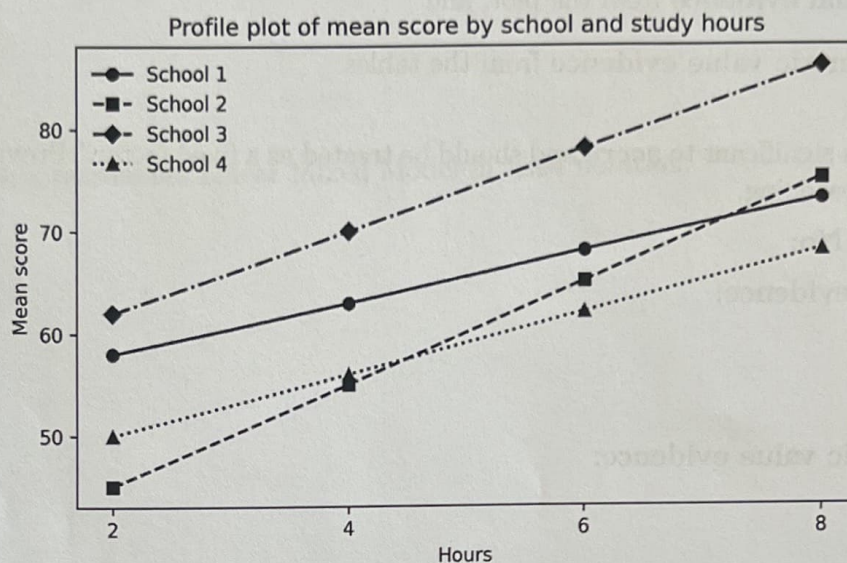
(e) (Exact LRT, no χ^2)

Find the exact rejection region about X_A, X_B at level $\alpha = 0.05$. State the general rejection rule about X_A, X_B (may not be 4 and 0 as in this question). Then determine whether $x_A, x_B = (4, 0)$ should be rejected.

Hint: Conceptually consider how $\Lambda(x_A, x_B)$ depends on $|x_A - x_B|$.

Question 4. (10 Points, 2.5/2.5/2/1/2 Points per part)

The following profile plot shows the mean score by school and study hours.



The corresponding numeric summary is:

School	Hours=2	Hours=4	Hours=6	Hours=8
1	58	63	68	73
2	45	55	65	75
3	62	70	78	86
4	50	56	62	68

First differences (increase per 2 hours):

School	$\Delta(4-2)$	$\Delta(6-4)$	$\Delta(8-6)$	Avg Δ
1	5	5	5	5
2	10	10	10	10
3	8	8	8	8
4	6	6	6	6

Based on the evidence above, answer the following questions and propose a reasonable LMM.

Important: For each subquestion (a)(b) below, you must provide:

- one **visual evidence** from the plot, and
- one **numeric value evidence** from the tables.

- (a) Is hours significant to score and should be treated as a fixed factor? Provide evidence-based reasoning.

Yes or No:

visual evidence:

numeric value evidence:

- (b) Should hours have a random slope by school? Provide evidence-based reasoning.

Yes or No:

visual evidence:

numeric value evidence:

- (c) Would you expect strong intercept-slope correlation? Briefly justify using evidence.

Yes or No:

One visual or numeric value evidence:

(d) Would you consider modeling heteroscedastic residual variance by hours? Give your reasoning.

(e) Propose a reasonable Linear Mixed Model in `lme4` notation.